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Sleep disturbance, cancer-related fatigue, and depression as determinants of quality of life among breast cancer patients undergoing chemotherapy: a cross-sectional study

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Abstract

Purpose This study aims to assess effect of sleep disturbance, cancer-related fatigue, and depression on the quality of life (QoL) of breast cancer patients undergoing chemotherapy.

Method A descriptive, cross-sectional study was conducted at an oncology hospital in Cairo, Egypt, from November 2024 to February 2025. A total of 253 breast cancer patients undergoing chemotherapy were recruited through convenience sampling. Data were collected using validated instruments: Pittsburgh Sleep Quality Index (PSQI), Functional Assessment of Cancer Therapy-Breast (FACT-B), Cancer Fatigue Scale (CFS), and Patient Health Questionnaire-9 (PHQ-9) for depression. Linear regression models were employed to analyze the relationships between sleep disturbance, fatigue, depression, and QoL.

Results Findings revealed that 99.2% of patients experienced fatigue, 87.4% reported poor sleep, and 93.3% exhibited depressive symptoms. Regression analysis demonstrated that sleep disturbance had the strongest negative impact on QoL ($B = -1.506, p < 0.001$), overshadowing both fatigue and depression. Correlation analysis further supported these findings, with sleep disturbance showing the highest negative correlation with QoL ($r = -0.81, p < 0.001$), followed by fatigue ($r = -0.72, p < 0.001$) and depression ($r = -0.56, p < 0.001$).

Conclusions Sleep disturbance, cancer-related fatigue, and depression are significantly associated with reduced QoL in breast cancer patients undergoing chemotherapy. Among these factors, sleep disturbance appears to be the most influential. Interventions targeting sleep quality may help mitigate related psychological and physical challenges, thereby enhancing patients' overall well-being.

Keywords Breast Cancer, Quality of life, Sleep disorder, Fatigue, Depression, Chemotherapy

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Introduction

Breast cancer is a leading health priority, being the most commonly diagnosed malignancy globally. In 2022, 2.3 million women were diagnosed with breast cancer, resulting in 670,000 fatalities worldwide. By 2050, the incidence of new cases and mortality rates will have risen by 38% and 68%, respectively [17, 33]. It is also the largest source of female cancer-related mortality [55]. A meta-analysis and systematic review in Egypt highlighted that breast cancer is the most frequent cancer among Egyptian women, accounting for 42% of cancers, and most patients come to the hospital at advanced stages [7]. Even though nearly 99% of breast cancer occurs in women, a minority (0.5–1%) also occurs in men. It may occur in women of any age group post-puberty, with the frequency rising with advancing age [53].

Advances in therapy have reduced breast cancer mortality dramatically and improved disease-free survival. However, attaining a satisfying quality of life (QoL) for patients remains difficult. The focus has now shifted towards measuring survivorship in regards to QoL, since cancer affects not just physical health but various aspects of a patient's life [32]. Health-related QoL is a valuable outcome measure for cancer care, which provides informative data for patients as well as clinicians. QoL measurement helps in setting realistic expectations regarding the impact of treatments on well-being and daily function, identifies common problems that must be treated, and helps to assess the impact of therapies and supportive care interventions [25].

Quality of life (QoL) refers to an individual's perception of their position in life, influenced by their cultural and value systems, personal goals, expectations, standards, and concerns [29]. It is often linked to overall life satisfaction and well-being. QoL is a multidimensional concept that includes physical health, psychological well-being—such as anxiety and depression levels—and social support [45].

Breast cancer is not solely a physical illness; it is also a condition deeply linked to psychosocial challenges [30]. Patients often experience a range of physical effects, including fatigue, pain, lymphedema, sexual dysfunction, and sleep disturbances [49]. Beyond these physical consequences, breast cancer can lead to psychological distress, manifesting as fear, anxiety, depression, and diminished life satisfaction [39]. The diagnosis and treatment process can intensify emotional distress, contributing to feelings of anxiety, depression, anger, uncertainty, and hopelessness [30]. These negative effects significantly impact both health and overall QoL [10]. Among the various treatment options for breast cancer, chemotherapy is particularly known for its severe side effects, which can profoundly affect a patient's QoL and influence their willingness to continue treatment [35].

Studies highlight the significant impact of sleep disturbance, depression, and cancer-related fatigue well beyond the treatment phase of cancer. These symptoms are some of the most powerful predictors of QoL because they play a role in chronic insomnia, fatigue, and emotional distress. Insomnia and fatigue are exceptionally common among patients with breast cancer, as among other patients with chronic illnesses [16]. In addition, these symptoms go underdiagnosed and undertreated among patients with cancer, with consequent prolonged suffering and suboptimal well-being [16].

A study on 372 patients with breast cancer undergoing chemotherapy revealed that 99.2% of them experienced fatigue, 87.4% reported poor sleep, and 93.3% exhibited depressive symptoms [26]. Another study also indicated that cancer fatigue has a significant impact on QoL [56]. This type of fatigue not only disrupts patients' normal functioning in performing daily activities and maintaining working but also contributes to the development of low mood and depression, eventually interfering with both their quality of life as well as the effectiveness of the rehabilitation interventions [52]. Moreover, another research conducted by Akin and Kas Guner [2] examined the interrelation of QoL with cancer-related fatigue. It validated that cancer-related fatigue effectively diminishes cancer patients' quality of life [2].

Depression is a significant health concern for breast cancer survivors and is associated with lower HRQOL [47]. In addition, two studies found that depression is correlated with a lower quality of life [23, 54]. Given the strong connection between these psychological and physical challenges, addressing sleep disturbances, fatigue, and depression is crucial for enhancing the well-being and recovery of breast cancer patients.

Cancer patients in Egypt encounter numerous psychosocial challenges that greatly affect their quality of life, similar to those in various other regions. A study at the Oncology Center of Mansoura University, which involved 175 cancer patients, revealed significant levels of psychological distress, including anxiety and depression [19]. A case-control study conducted at the Minia Oncology Center, involving 400 participants, found significantly higher levels of moderate to severe depression, anxiety, and stress among cancer patients, underscoring the critical need for regular psychological screening in the cancer care [48]. Additionally, research involving 550 individuals recently diagnosed with cancer indicated that 46% faced significant distress, especially among those in the later stages of the disease. Major sources of distress encompass decisions related to treatment, personal issues, and anxiety about the future [1]. In a similar vein, a study with 200 women receiving treatment for breast cancer revealed a significant link between psychological distress and unmet supportive care needs [18].

While previous research has established individual links between sleep disturbances, fatigue, depression, and QoL, few studies have examined their combined effects using multivariate analysis. So, this study seeks to bridge this gap by analyzing how these three factors collectively influence QoL in breast cancer patients undergoing chemotherapy. By using linear regression models, this study aims to determine which factor exerts the greatest impact on QoL. Understanding these relationships will provide critical insights for healthcare providers, enabling them to develop targeted interventions that address not only cancer treatment but also the broader well-being of patients.

This study aims to assess the effect of cancer-related fatigue, sleep disturbance, and depression on the QoL in breast cancer patients undergoing chemotherapy. By evaluating these factors, the study seeks to identify the strongest predictors of QoL, offering evidence to guide targeted interventions that can improve the overall well-being of patients during treatment.

Hypotheses

- H₁: Depression has a significant negative effect on the quality of life in breast cancer patients undergoing chemotherapy.
- H₂: Cancer-related fatigue has a significant negative effect on the quality of life in breast cancer patients undergoing chemotherapy.
- H₃: Sleep disturbance has a significant negative effect on the quality of life in breast cancer patients undergoing chemotherapy.
- H₄: Sleep disturbance, cancer-related fatigue, and depression together explain a substantial proportion of the variation in QoL.
- H₅: Patients with a longer duration since diagnosis report higher levels of depression compared to those more recently diagnosed.

Methods

Research design

A descriptive cross-sectional study using a self-report questionnaire was conducted from November 2024 to February 2025 and reported according to the guidelines for Strengthening the Reporting of Observational Studies in Epidemiology (STROBE).

Study setting

This study was conducted at an oncology hospital, a governmental facility in Cairo, Egypt, affiliated with the Ministry of Health, which provides various medical services, including chemotherapy, to cancer patients.

Sample size

To calculate the required sample size for the study, we applied the T-test Independent Power formula, a widely used approach to determining sample size by effect size, statistical power, and significance level [50]. Employing an effect size of -0.388 from a prior study [51], 85% power, and 0.05 significance level, we estimated a minimum sample size of approximately 120 participants. Because there are multiple predictors sleep disturbance, cancer-related fatigue, and psychological distress—while there is just one outcome variable (quality of life), we had a potential 15% rate of attrition in mind. This re-weighting increased the required sample size to about 138 patients in order to cover for missing data and dropouts. However, in case of complex statistical models, such as latent growth modeling or structural equation modeling, a larger sample size (usually ranging from 200 to 300) is recommended in order to enhance the validity of the results [42]. The ultimate sample size was thus set at 253 patients.

Participants

Participants were recruited using a convenience sampling method, targeting women diagnosed with breast cancer at the designated hospital. Eligible patients were identified through clinical consultation and review of medical records; selection was non-random and based on patient availability and willingness to participate during scheduled chemotherapy visits. This approach was adopted due to practical constraints, including limited time, resources, and patient flow during the data collection period. It enabled efficient access to patients while maintaining feasibility in reaching the required sample size. The inclusion criteria were that the participants were females aged 18–65 years, diagnosed with breast cancer, undergoing chemotherapy, with normal mental and cognitive status, and able to cooperate throughout treatment, follow-up, and completion of questionnaires. The exclusion criteria were a history of recurrence of breast cancer “Patients with recurrent disease may have different psychological and physical symptom profiles, which could confound the results and reduce comparability with those undergoing first-time treatment”, other concurrent primary cancers “The presence of other cancers could introduce additional sources of fatigue, depression, or sleep disturbance unrelated to breast cancer or its treatment.”, and pregnancy “Pregnancy could independently affect sleep quality, fatigue, mood, and quality of life, making it difficult to isolate the effects of cancer and chemotherapy”.

Eligible patients were approached in person at the outpatient chemotherapy unit during their scheduled treatment visits. After assessing eligibility through consultation with healthcare providers and reviewing

medical records, the research team extended verbal invitations to 350 patients. The study's purpose and procedures were explained in detail, and informed consent was obtained on-site from 307 patients, resulting in a response rate of 87.7%. Of those, 18 patients withdrew before data collection, 16 did not complete the interview, and 20 were excluded due to missing data. Consequently, the final sample analyzed consisted of 253 patients.

Data collection instruments

Patient interview questionnaire

Characteristics of nurses, such as age, Marital status, Employment status, Cancer stage, and Months since diagnosis [31].

Sleep disturbance

The Pittsburgh Sleep Quality Index (PSQI), developed by Buysse et al. [9], was utilized to evaluate sleep disturbances. This 19-item instrument measures seven key components of sleep quality: subjective sleep quality, sleep latency, sleep duration, sleep efficiency, sleep disturbances, use of sleep medication, and daytime dysfunction. The total sleep quality score is derived from these seven components, ranging from 0 to 21. Each element is rated on a scale from 0 (no difficulty) to 3 (severe difficulty), with higher scores indicating more significant sleep disturbances. A global PSQI score above 5 is generally considered indicative of poor sleep quality [22]. The Arabic version of the PSQI has demonstrated reliability, with a Cronbach's alpha coefficient of 0.77 [3]. Additionally, a previous study reported a Cronbach's alpha value of 0.89 for the scale [51].

Quality of life

The Quality of life was measured using the Functional Assessment of Cancer Therapy-Breast (FACT-B), it was developed by Cella [12], a five subscale questionnaire with 37 items: physical well-being (range 0–28), social/family well-being (range 0–28), emotional well-being (range 0–24), functional well-being (range 0–28), and a breast cancer-specific subscale (range 0–40). Each is scored on an ordinal scale with 5 items ranging from 0 to 4, such that the total score ranges from 0 to 144, in which higher is better. The FACT-B has demonstrated superior internal consistency, and the Cronbach's alpha was 0.951 [51].

Cancer fatigue scale

The scale was employed to evaluate fatigue in cancer patients, offering insights into its impact on their physical, emotional, and cognitive health, it was developed by Okuyama et al. [43]. It has 15 items and three subscales: physical (7 items), affective (4 items), and cognitive (4 items). All the items are scored from 1 to 5, and the range

of total scores is from 15 to 75. Higher scores on the CFS indicate greater fatigue, which may interfere with the functioning and engagement in daily activities by the patient. In another research, the scale was highly consistent internally, as shown by Cronbach's alpha of 0.88 [43].

Depression- PHQ-9

The PHQ-9 is a nine-item tool used to screen for depression. It was developed by Kroenke et al. [34]. It requires patients to reflect on how frequently they experienced each symptom over the past two weeks. Response options range from "not at all" (0) to "nearly every day" (3). The total score spans from 0 to 27, with thresholds of ≥ 5 , ≥ 10 , and ≥ 15 corresponding to mild, moderate, and severe depression, respectively [34]. The PHQ-9 has been shown to have strong reliability, with a Cronbach's alpha of ≥ 0.84 [27].

Data collection

Structured interview questionnaires were conducted with the studied patients to collect data for this study. This process took place from November 2024 to February 2025. Patients were identified through the hospital's oncology outpatient chemotherapy unit. Eligible participants were screened based on their medical records, which included documented diagnosis of breast cancer, treatment status, and clinical eligibility confirmed by the attending oncologist. Details about the study's goals, possible advantages, and participants' rights were explained to the patients before participating in the study. We divided the patients into small groups to facilitate conducting the interview for collecting the data as characteristics of patients, Sleep disturbance, Quality of life, Cancer fatigue scale, and Depression- PHQ-9. All structured interviews were conducted in the outpatient chemotherapy unit of the oncology hospital. Patients were approached and interviewed in a private area within the unit while waiting for or immediately after receiving their scheduled chemotherapy sessions. This setting provided a familiar, non-intrusive environment and ensured patient comfort and privacy during data collection. The researcher was available throughout the data collection period to address any questions, ensuring participants fully understood the study's objectives. Patients who participated in the pilot study were included in the primary study sample. Finally, incomplete questionnaires were excluded, and any missing data were corrected before analysis.

Validity and reliability

The researchers evaluated the reliability of the study instruments by employing Cronbach's alpha coefficient test with a statistical significance threshold of $p < 0.05$. Tool 1 (Pittsburgh sleep quality Index (PSQI)) and Tool

Table 1 Characteristics of studied breast cancer patients ($n = 253$)

Characteristics	no.	%
Age (years)		
Mean (SD)	46.3 (11.7)	
Marital status		
Single	77	30.4
Married	140	55.3
Divorced	24	9.5
Widow	12	4.8
Employment status		
Employee	119	47
Unemployed	88	34.8
Retired	46	18.2
Cancer stage		
I	52	20.6
II	71	28.1
III	85	33.6
IV	45	17.7
Months since diagnosis		
Mean (SD)	32.52 (15.6)	

SD Standard Deviation

2 (Functional Assessment of Cancer Therapy-Breast (FACT-B)), Tool 3 (Cancer fatigue scale), and Tool 4 (Depression- PHQ-9) exhibit reliability, surpassing the permissible cutoff of ≥ 0.7 for study groups' values, with $\alpha = 0.819, 0.862, 0.813$ and 0.846 , respectively. Pilot testing of the survey instruments with a representative sample was conducted to assess the items for intelligibility and practicality, identify potential obstacles and concerns during data collection, and test the time required to complete the tools. Pilot study participants were included in the study sample.

Statistical analysis

Statistical Package for Social Sciences (SPSS) was used to code and enter the collected data. Descriptive statistics were used to assign demographic characteristics into numbers, that is, frequency distributions, means, standard deviations, and percentages. Quantitative data were presented as values and percentages. The Kolmogorov–Smirnov test was used to test the normality of the data distribution. A correlation heatmap was generated to visually display correlations between sleep disturbance, quality of life (QoL), cancer-related fatigue, and depression. A longitudinal analysis was conducted to compare the correlations between QoL and key health indicators in patients with breast cancer. To explore these correlations better, a Python script was used to generate a correlation heatmap using synthetic longitudinal data. The script employed Pandas, NumPy, and Seaborn/Matplotlib to manipulate and visualize the data. Furthermore, linear regression analysis was employed to assess the relationship between QoL and significant independent

Table 2 Mean scores of depression, cancer fatigue, quality of life, and sleep disturbance ($n = 253$)

Study Variables	Mean	SD	Min	Max	Interpretation
Depression	11.58	2.86	6	19	Moderate
Cancer fatigue	47.9	8.03	35	68	High
Quality of life	58.11	7.29	43	81	poor
Sleep disturbance	11.05	4.57	4	19	Poor sleep

SD Standard Deviation

variables, such as sleep disturbance, depression, and cancer fatigue. Statistical significance was set at $P < 0.05$.

In order to evaluate possible selection bias stemming from participant drop out, we performed sensitivity analysis of the 253 patients completers compared to the 54 patients non-completers (withdrawn, incomplete interviews, or missing data). A set of independent samples t-tests and chi-square tests were performed for each and compared for within group differences: age, cancer stage, employed, marital status and months since diagnosis. The analysis did not find statistically significant differences in age ($p = 0.66$), cancer stage ($p = 0.98$), employment status ($p = 0.85$), months since diagnosis ($p = 0.57$), Marital Status ($p = 0.90$), see Table 1 in appendix. Based on these results, the final sample is likely representative of the original group, and attrition is unlikely to cause systematic bias on study outcomes.

Results

Table 1 presents the sociodemographic and clinical characteristics of 253 patients with breast cancer in this study. The mean age of the participants was 46.3 years ($SD = 11.7$). The majority were married (55.3%), followed by single (30.4%), divorced (9.5%), and widowed (4.8%). Regarding employment status, nearly half of the participants were employed (47%), 34.8% were unemployed, and 18.2% were retired. The distribution of cancer stages showed that the largest proportion of patients was in Stage III (33.6%), followed by Stage II (28.1%), Stage I (20.6%), and Stage IV (17.7%). Additionally, the mean time since diagnosis was 32.52 months ($SD = 15.6$), indicating variability in the time since diagnosis.

The mean score for depression was 11.58 ± 2.86 , with values ranging from 6 to 19, indicating moderate depression. The mean Cancer fatigue was 47.9 ± 8.03 , with a range from 35 to 68, indicating a higher level of fatigue. The mean Quality of life mean was 58.11 ± 7.29 , with a range from 43 to 81, indicating a poor quality of life. The mean sleep disturbance was 11.05 ± 4.57 , with a range from 4 to 19 years, indicating poor sleep (Table 2).

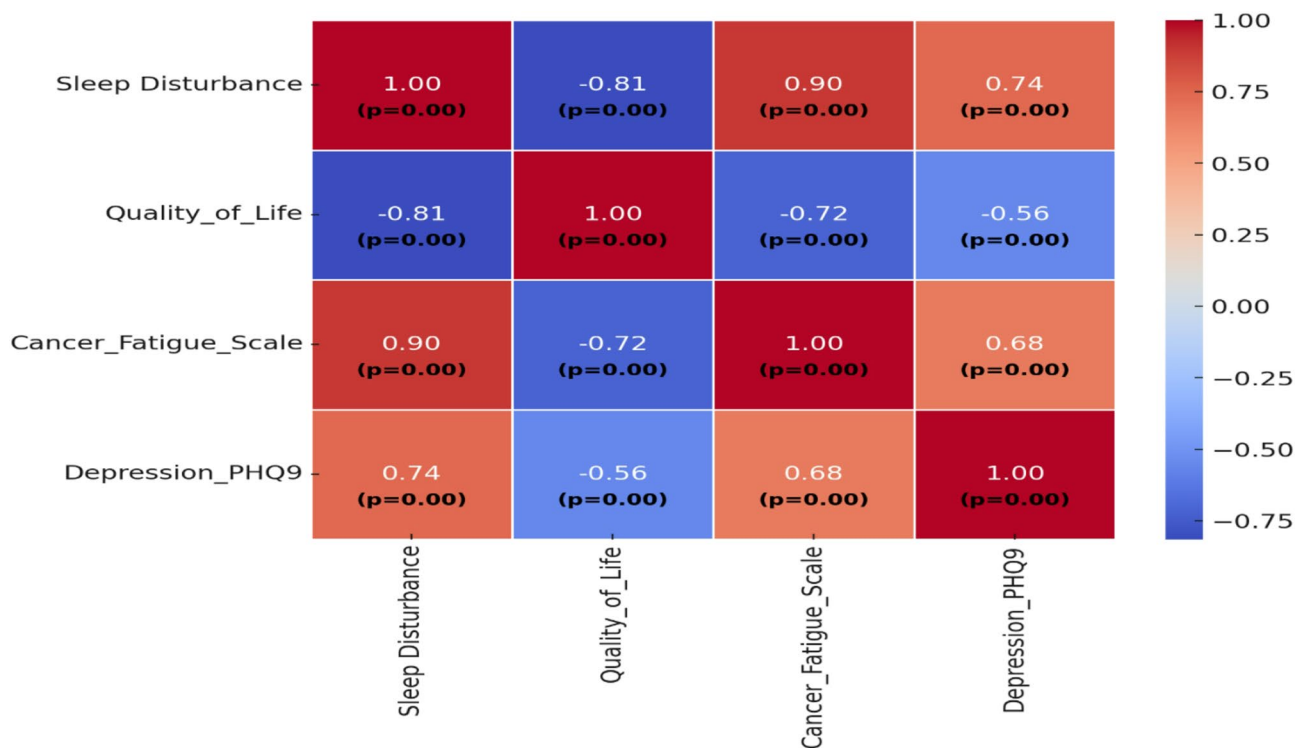
According to the linear regression for quality of life of breast cancer patient (Table 3), Model 1 showed that depression ($B = -1.423, p = 0.000$) had a significant adverse effect on quality of life, explaining 31.2% of the variation in quality of life ($F = 113.9, p < 0.001$). In model 2, cancer

Table 3 Linear regression for quality of life of breast cancer patients ($n = 253$)

Model 1				Model 2			Model 3		
Variable	B	SE	p	B	SE	p	B	SE	p
Intercept	74.61	1.5	0.000	89.31	1.9	0.000	68.97	2.53	0.000
Depression	-1.423	0.13	0.000	-0.347	0.150	0.000	-0.236	0.138	0.087
Cancer fatigue				-0.567	0.053	0.000	-0.163	0.075	0.399
Sleep disturbance							-1.506	0.144	0.000
	$R^2=0.312$, $F=113.9$, p 0.000			$R^2=0.524$, $F= 137.67$, p 0.000			$R^2=0.668$, $F= 167.44$, p 0.000		

F, p: f and p values for the model

B Unstandardized Coefficients, Beta Standardized Coefficients, t t-test of significance, SE Standard error

*Statistically significant at $p \leq 0.05$ **Fig. 1** Correlation heatmap for sleep disturbance, quality of life, cancer fatigue, and depression

fatigue was entered ($B = 0.567$, $P = 0.000$). and depression ($B = -0.347$, $p = 0.000$). The model explained 52.4% of the variation in the quality of life ($F = 137.67$, $p < 0.001$). In Model 3, Sleep disturbance ($B = 1.506$, $P = 0.000$), cancer fatigue ($B = -0.163$, $p = 0.399$), and depression ($B = -0.236$, $p = 0.087$) were included. The model explained 66.8% of the variation in the quality of life ($F = 167.44$, $p < 0.001$). These findings suggest that sleep disturbance exerts the most substantial impact on QoL, and when included, may overshadow the effects of fatigue and depression.

The results are shown in Fig. 1. highlighted the intricate relationships between fatigue, depression, sleep disturbance, and quality of life in breast cancer patient, which indicated a negative correlation between sleep disturbance and quality of life ($r = -0.81$, $p < 0.001$), fatigue and quality of life ($r = -0.72$, $p < 0.001$), depression, and

quality of life ($r = -0.56$, $p < 0.001$). There was also a positive correlation between sleep disturbance and depression ($r = 0.74$, $p < 0.001$), fatigue, and depression ($r = 0.68$, $p < 0.001$). Furthermore, there was a positive correlation between sleep disturbances and fatigue ($r = 0.90$, $p < 0.001$).

Sleep Disturbance (PSQI) vs. Quality of Life (Orange Line). The negative correlation between the PSQI and quality of life remained consistent regardless of time since diagnosis. This suggests that poor sleep quality consistently lowers quality of Life regardless of how long the patient has been diagnosed, indicating that sleep disturbances remain a long-term issue. Additionally, the Cancer Fatigue Scale vs. Quality of Life (Blue Line). Fatigue has the strongest negative correlation with Quality of Life, meaning that higher fatigue levels lead to significantly

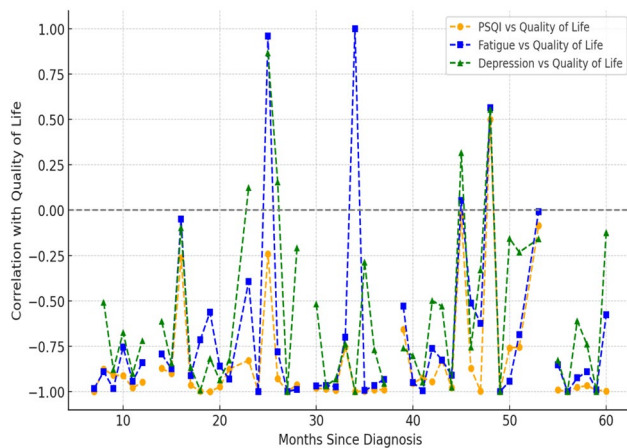


Fig. 2 Correlations between quality of life and key health factors in breast cancer patients

lower Quality of Life, showing that fatigue is a persistent issue affecting patients throughout their cancer journey. The magnitude of the correlation suggests that fatigue may be the most important predictor of Quality of Life in this dataset. Finally, the impact of depression (PHQ-9) on quality of life (represented by the green line) increases with time since the initial diagnosis. See more in Fig. 2.

Discussion

This study explores the impact of cancer fatigue, sleep disturbances, and depression on QoL among breast cancer patients undergoing chemotherapy. By assessing these variables, our research attempts to identify the best predictors of QoL, providing feedback that can inform targeted interventions towards enhancing patient wellness. While other studies have previously reviewed cyclic changes in fatigue, depression, sleep, and activity in cancer patients undergoing chemotherapy, this study is among the first to review how differences in these factors impact QoL compared with time since diagnosis.

The study findings revealed moderate level of depression (M 11.58 SD 2.86) potentially due to the psychological impact of cancer diagnosis, uncertainty about the future, and physical burden of chemotherapy, high level of cancer-related fatigue (M 47.9 SD 8.03) that reflects profound physical exhausting, likely exacerbated by chemotherapy's side effects, significantly impairing patients' ability to manage routine activities effectively, poor quality of life (M 58.11 SD 7.29) highlight how treatment related symptoms negatively influence patients' physical, emotional, and social well-being, reducing their capacity to engage fully in daily life and rehabilitation program.

Moreover, significant sleep disturbances (M 11.05 SD 4.57) underline the severity of insomnia and disturbed sleep patterns among breast cancer patients ($n=253$) undergoing chemotherapy, potentially amplifying their fatigue and depressive symptoms. These results

collectively indicate substantial physical, emotional, and psychological burdens affecting patients' overall well-being during their treatment.

These findings agree with previous studies that reported similar psychosocial problems in patients with breast cancer undergoing chemotherapy. For instance, it has been reported that prevalence of moderate to severe fatigue during chemotherapy ranges from 26 to 60% [6], depressive symptoms are experienced by 20–39% of patients [40], insomnia is self-reported by 79% [44], and inactivity has been self-reported in 40–74% [8]. Furthermore, depression and anxiety were also determined to account for 18.8% and 12.8% of the variance in sleep quality, respectively [59]. Cho and Hwang [14] reported that the average anxiety score was 6.89 out of 21, and 15.0% of the patients possessed a moderate level of anxiety (above 11 points). Similarly, the average depression score was 6.60 out of 21, and 15.8% were diagnosed with a moderate level of depression.

Follow-up studies have shown that the occurrence of fatigue is as high as 46.3% [24], and worsening fatigue during both chemotherapy regimens [36]. However, outcomes are contrary to those of Yfantis et al. [57], which showed that the overall composite score for the functional scales of the QLQ-C30 illustrated a fairly good QoL and low-grade symptomatology following surgery. Also, research on university hospital in-patients has found higher quality care and lower mortality rates [5].

In our study, we conducted three regression models to determine how these factors contribute individually and collectively to QoL. In Model 1, the single variable of depression accounted for 31.2% of the variability in predicting QoL. This shows that extra depressive symptoms considerably decrease a patient's perception of their QoL. In Model 2, both fatigue and depression were found to be significantly associated with QoL when cancer-related fatigue was added, which means that they could explain 52.4% of the variation. This reveals that fatigue plays a larger role in the impairment of QoL than depression, and both have a cumulative effect. Model 3, which focused on sleep disturbance, was added to the analysis, and the model confirmed a large part of the variance ($R^2 = 0.668$). In this full model, sleep disorders were shown to have the most intense negative link with QoL ($B = -1.506$, $p < 0.001$). It should be noted that the effects of both cancer fatigue ($B = -0.163$, $p = 0.399$) and depression ($B = -0.236$, $p = 0.087$) were no longer statistically significant once sleep disturbance was factored in. These results support sleep disturbance as the main factor, which is more important than the combined effects of fatigue and depression on QoL. Notably, the involvement of sleep deficiency was the most influential factor in the predictive model; its contribution substantially affected both depression and fatigue. Thus, poor sleep quality may

be a central factor aggravating physical and psychological distress in patients.

The findings derived from the correlation heatmap further confirm the outcomes mentioned above and depict very negative interrelations between QoL and sleep disturbance ($r = -0.81$), fatigue ($r = -0.72$), and depression ($r = -0.56$). In addition, negative associations could also be observed between sleep disturbance, depression ($r = 0.74$), and fatigue ($r = 0.90$) at p value < 0.01 , the latter emphasizes the combined nature of these conditions and also suggesting that these problems do not just appear separately but rather reinforce each other. To sum up, attending to sleep problems seems imperative in raising the life quality of breast cancer patients with chemotherapy. Interventions directed to improving sleep quality may lead to the relief of related symptoms like depression and fatigue, therefore putting a patient's overall well-being on a more solid ground.

These findings are supported by various studies. For instance, Reyes-Gibby et al. [47] reported that depression was inversely related to HRQOL subscales for functioning, financial well-being, and general health, but was directly related to symptom severity. Similarly, studies have shown a negative relationship between depression and QoL [13]. Emre and Yilmaz [20] also observed that patients with breast cancer who reported poor sleep quality and high anxiety and depression were found to have a significantly worse quality of life, and such conditions further impaired their well-being.

In addition, research concerning the relationship between subjective sleep quality and QoL has all equated sleep disorders with lower QoL. This relationship has been observed in patient populations and control populations [36, 37, 38]. Regression analyses of anxiety and depression, with sleep quality as a factor, were significant. Of the sleep quality factors, daytime dysfunction explained the most for anxiety [14]. Also, low sleep quality, depression, and lower scores in QoL were strongly correlated with each other among breast cancer patients [46, 47, 58].

Furthermore, Al-Sharman et al. [4] found that sexual dysfunction, low sleep quality, depression, and anxiety were the predictors of QoL as identified by multiple regression analysis ($p \leq 0.05$). Evidence in support of these results comes from research that also establishes depression deteriorates sleep quality [21]. Also, trends on disturbed sleep and tiredness related to cancer showed increasing was followed immediately by steady recuperation, whereas trends on QoL and psychological distress also showed a trend toward diminished effect as time passed on [51]. According to these findings, follow-up care and early psychological interventions should be implemented at the initial diagnostic stage by physicians

to prevent high depression among breast cancer patients, ultimately improving their QoL [28, 41].

Poor sleep quality, fatigue, and depression all negatively impact QoL in cancer patients. Sleep disturbances remain a long-term issue, consistently reducing QoL. Fatigue has the most substantial negative effect, making it a key predictor of well-being. Depression tends to worsen in patients who diagnosed for a longer duration than newly diagnosed, playing an increasingly significant role in diminishing quality of life as patients progress through their cancer journey. It remains a persistent challenge and may intensify during extended cancer treatment. Addressing these factors collectively is crucial for improving patient outcomes.

These findings are in agreement with previous research, such as the study of Reyes-Gibby et al. [47], which followed the prevalence of depression over time with a general decline trend as years since treatment increased. Longitudinal variation has also been observed in the profound variations in fatigue, depression, sleep disturbances, and activity level. Fatigue increases, for instance, were strongly correlated with increases in depression prior to chemotherapy infusions [31]. Furthermore, it has been suggested that the prevalence of depression decreases with greater time since treatment [11, 15].

Moreover, although the attrition rate for our study was 17.6%, we performed a sensitivity analysis comparing baseline traits for completers and non-completers. The groups did not differ statistically, thus suggesting selection bias is likely of little consequence. Still, we recognize this as a limitation and propose that future research employs methods to minimize attrition and enhance validation through prospective tracking aiming for greater representativeness.

Although the associations identified in this study align with findings from international research, the present study offers important contributions within the Egyptian context. Breast cancer patients in Egypt often present at more advanced stages, face limited access to specialized psychosocial services, and may experience cultural stigma surrounding mental health. By documenting the significant impact of sleep disturbance, fatigue, and depression on quality of life in this setting, the study highlights critical gaps in supportive care and underscores the need for integrating mental health and symptom management into routine oncology services. These findings can support evidence-based recommendations for local clinical practice, inform public health priorities, and guide the development of culturally appropriate psychosocial interventions in cancer care.

Limitation

While the study provides valuable insights, it is limited in that it relied on self-reporting, which may result in

response bias. So, we suggested that further study should incorporate objective assessments of sleep patterns and fatigue levels to enhance accuracy. Additionally, the use of convenience sampling may introduce selection bias and limit the generalizability of the findings. However, to mitigate this concern, we performed a sensitivity analysis comparing completers (253 patients) and non-completers (54 patients) who excluded from the analysis on key demographic and clinical variables. The absence of statistically significant differences suggests that the final sample is broadly representative of the initial population approached for participation.

Conclusion

Our study highlights the triple burden of cancer-related fatigue, sleep disturbance, and depression on the QoL among breast cancer patients undergoing chemotherapy. In addition, sleep disturbance had the most substantial impact on QoL, while fatigue remains a persistent challenge, and depression becomes more critical with patients who have been diagnosed for a longer duration than newly diagnosed. Comprehensive, multidisciplinary interventions targeting these factors are essential for enhancing patient well-being throughout their cancer journey.

Clinical implications

Sleep interventions are essential in improving QoL. Cognitive-behavioral therapy for insomnia and relaxation techniques may be beneficial. Fatigue management strategies, including energy conservation, physical activity, and nutritional support, are also essential. Counseling, support groups, and antidepressant therapy may help mitigate these effects. A multidisciplinary approach addressing fatigue, sleep quality, and mental health simultaneously may yield the most significant improvements in patients' QoL.

Abbreviations

PSQI	Pittsburgh sleep quality index
FACT-B	Functional assessment of cancer therapy-breast
CFS	Cancer fatigue scale
PHQ-9	Patient health questionnaire-9
STROBE	Strengthening the reporting of observational studies in epidemiology

Supplementary Information

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Supplementary Material 1.

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Authors' contributions

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Data availability

All relevant data are available within the manuscript.

Declarations

Ethics approval and consent to participate

Our study was submitted and approved by local ethics committee at MTI university, reference number FAN/152/2024. All participating patients were provided comprehensive information about the study's purpose, objectives, and potential benefits. The researchers emphasized the study's voluntary nature, and patients could withdraw their participation without facing any consequences. Patients were required to provide written informed consent before participating in the study. Participation was entirely voluntary, and patients had the right to withdraw at any stage without any consequences. The collected data was coded to maintain confidentiality, ensuring no identifiable information was disclosed. Our study adhered to the Declaration of Helsinki principle.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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